

Design and Development of Banking Management System

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Mini Project Report – CS513 System Software

Abstract

The Banking Management System is a client-server-based software designed to simulate the core functionalities of a bank. It provides secure access to customers, employees, managers, and administrators, supporting operations such as deposits, withdrawals, fund transfers, loan management, and feedback handling. The system emphasizes concurrency, synchronization, and data consistency through Linux system calls and socket programming.

Objectives

- 1 To design a modular banking management system using system-level programming concepts.
- 2 To provide concurrent client access to shared banking data through socket-based communication.
- 3 To ensure data consistency using file management, locking mechanisms, and synchronization.
- 4 To simulate a real-world banking environment with multiple user roles.

System Design Overview

The project follows a modular client-server architecture. The server maintains all banking data and handles multiple clients concurrently through socket programming. Each client connects via a terminal interface to perform operations based on their role. File locking and synchronization mechanisms prevent race conditions when multiple clients access shared data.

Module Descriptions

Customer

- 1 Login and view account balance.
- 2 Deposit, withdraw, and transfer funds securely.
- 3 Apply for loans and view transaction history.
- 4 Provide feedback and change password.

Employee

- 1 Add and modify customer details.
- 2 Process loan applications and view transactions.
- 3 Change password and manage assigned loan tasks.

Manager

- 1 Activate or deactivate customer accounts.
- 2 Assign loan processing tasks to employees.
- 3 Review feedback and oversee branch operations.

Administrator

- 1 Add new employees and manage user roles.
- 2 Modify customer and employee details.
- 3 Perform high-level administrative tasks.

Implementation Details

The implementation primarily uses C with Linux system calls for process and file management. The server uses ``socket()``, ``bind()``, ``listen()``, and ``accept()`` to handle multiple clients concurrently. Clients connect using ``connect()`` and communicate through ``read()`` and ``write()`` calls. Critical sections involving file operations employ ``fcntl()`` locks to ensure atomic updates.

Results

The Banking Management System successfully simulates multi-user access to a bank's database. Multiple clients can log in, perform transactions, and manage accounts simultaneously without data corruption. The system demonstrates proper use of concurrency, synchronization, and secure data handling in a multi-process environment.

Conclusion and Future Scope

The Banking Management System fulfills its objectives of demonstrating concurrent file handling, secure access control, and inter-process communication using system-level programming. Future improvements could include integrating database support, data encryption, and GUI-based access for better usability.